TWO CONSECUTIVE ADMISSIBLE MONOMORPHISMS OF DG MODULES GIVE TRIANGLES SATISFYING TR4

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Editorial note. The counted statement is the author's own lemma environment labelled lemma-two-split-injections (source lines 1285-1292); the proof environment begins two lines below it and is its own proof, closing at source line 1341.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; the notions of dg algebra, admissible monomorphism and TR4 are the author's own surrounding development and are named rather than cited by label.

Editorial note. The proof contains two displayed diagrams written with the plain matrix environment and one displayed triangle; both are part of the retained source range and are reproduced as drawn, so no figure file is needed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: two composable admissible monomorphisms of dg modules give three distinguished triangles satisfying TR4.

Allowed standard prerequisites (editorial).

- Differential graded algebra and dg module. A differential graded algebra A is a graded algebra with a differential of degree one satisfying the Leibniz rule, and a dg A-module is a graded module with a compatible differential.. It is the ambient category in which all the modules, morphisms and shifts live (source lines 1286-1288 and throughout the proof).
- Admissible monomorphism and its cokernel. A morphism of dg modules is admissible when it fits into a short exact sequence of graded modules that splits as graded modules; its cokernel then carries an induced differential making the cone a distinguished triangle.. The hypothesis makes alpha and beta admissible, which produces the left inverses, the cokernels Q_i and hence the distinguished triangles (source lines 1295-1300, 1316-1318).
- Axiom TR4 (octahedral axiom). Given distinguished triangles (K,L,Q_1), (L,M,Q_3) and (K,M,Q_2) built from two composable morphisms, TR4 asserts that the connecting morphism Q_3 -> Q_1[1] can be chosen so that the three triangles form the octahedral diagram.. It is the conclusion: the proof produces exactly such a structure at source lines 1335-1340.

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Lemma 0.1. *Let (A, d) be a differential graded algebra. Suppose that $\alpha : K \to L$ and $\beta : L \to M$ are admissible monomorphisms of differential graded A-modules. Then there exist distinguished triangles $(K, L, Q_1, \alpha, p_1, d_1)$, $(K, M, Q_2, \beta \circ \alpha, p_2, d_2)$ and $(L, M, Q_3, \beta, p_3, d_3)$ for which TR4 holds.*

Proof. Say $\pi_1 : L \to K$ and $\pi_3 : M \to L$ are homomorphisms of graded A-modules which are left inverse to α and β. Then also $K \to M$ is an admissible monomorphism with left inverse $\pi_2 = \pi_1 \circ \pi_3$. Let us write Q_1, Q_2 and Q_3 for the cokernels of $K \to L$, $K \to M$, and $L \to M$. Then we obtain identifications (as graded A-modules) $Q_1 = \mathrm{Ker}(\pi_1)$, $Q_3 = \mathrm{Ker}(\pi_3)$ and $Q_2 = \mathrm{Ker}(\pi_2)$. Then $L = K \oplus Q_1$ and $M = L \oplus Q_3$ as graded A-modules. This implies $M = K \oplus Q_1 \oplus Q_3$. Note that $\pi_2 = \pi_1 \circ \pi_3$ is zero on both Q_1 and Q_3. Hence $Q_2 = Q_1 \oplus Q_3$. Consider the commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & K & \to & L & \to & Q_1 & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & K & \to & M & \to & Q_2 & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & L & \to & M & \to & Q_3 & \to & 0 \end{array}$$

The rows of this diagram are admissible short exact sequences, and hence determine distinguished triangles by definition. Moreover downward arrows in the diagram above are compatible with the chosen splittings and hence define morphisms of triangles

$$(K \to L \to Q_1 \to K[1]) \longrightarrow (K \to M \to Q_2 \to K[1])$$

and

$$(K \to M \to Q_2 \to K[1]) \longrightarrow (L \to M \to Q_3 \to L[1]).$$

Note that the splittings $Q_3 \to M$ of the bottom sequence in the diagram provides a splitting for the split sequence $0 \to Q_1 \to Q_2 \to Q_3 \to 0$ upon composing with $M \to Q_2$. It follows easily from this that the morphism $\delta : Q_3 \to Q_1[1]$ in the corresponding distinguished triangle

$$(Q_1 \to Q_2 \to Q_3 \to Q_1[1])$$

is equal to the composition $Q_3 \to L[1] \to Q_1[1]$. Hence we get a structure as in the conclusion of axiom TR4. $\square$